

Line follower circuit architecture

Common
Ground
among all

This circuit is made considering that rls 08 sensor runs fine at 3.3v supply, if not it will be powered by 5v supply from the buck converter and 8 voltage dividers will be used to map out voltage from analog pins to a logic level of 3.3V.

- 2S LIPO Battery pack
- 7.4v
- 850 mAh
- 25C , 50C (Burst)
- Max discharge - 21A



(7.4v rail)

Power Switch

max -230mA current draw

Buck regulator - LM2596
(7.4V to 5v)

- Continuous current rating - 2A
- Max current draw - 3A

(5v rail)
max - 500mA approx current draw

ESP32 Vin pin/
5v Supply

(3.3V LDO rail from esp32)
Max - 150mA current draw

- TB6612FNG VCC
- RLS-08 Array VCC

N20 motors
encoders pins
VCC pin logic

ESP32 GPIOs
(encoders:
13,14,27,26)

Motor Driver (Vm)
(TB6612FNG)

- Ratings per channel - Peak Current: 3.2A , Avg Current: 1.2A
- Vmax(motor) - 15v
- Logic supply - 2.7-5v
- Logic high Vin = $0.7 \cdot V_{CC}$

Voltage divider (for
battery voltage
detection)

(0-2.8v)

ESP 32 GPIO 25
(ADC)

N20 Motors
Rating: 6V, 200RPM
no load current - 30mA
Rated current - 60mA
Stall current - 230mA

Direct 3.3V PWM ,Direction Controls to
TB6612FNG , (pin: 16,17,18,21,22,23)

Direct 3.3V Analog Inputs from RLS-08:
Pins 32-39 (ADC 1)

push button input , status led output
(pins: 2,4,5)